

## Data Flow Diagram

|               |                                             |
|---------------|---------------------------------------------|
| Date          | 10 JULY 2026                                |
| Team ID       |                                             |
| Project Name  | Emotion Detection & Learning Support Engine |
| Maximum Marks | 2 Marks                                     |

### Data Flow Diagram

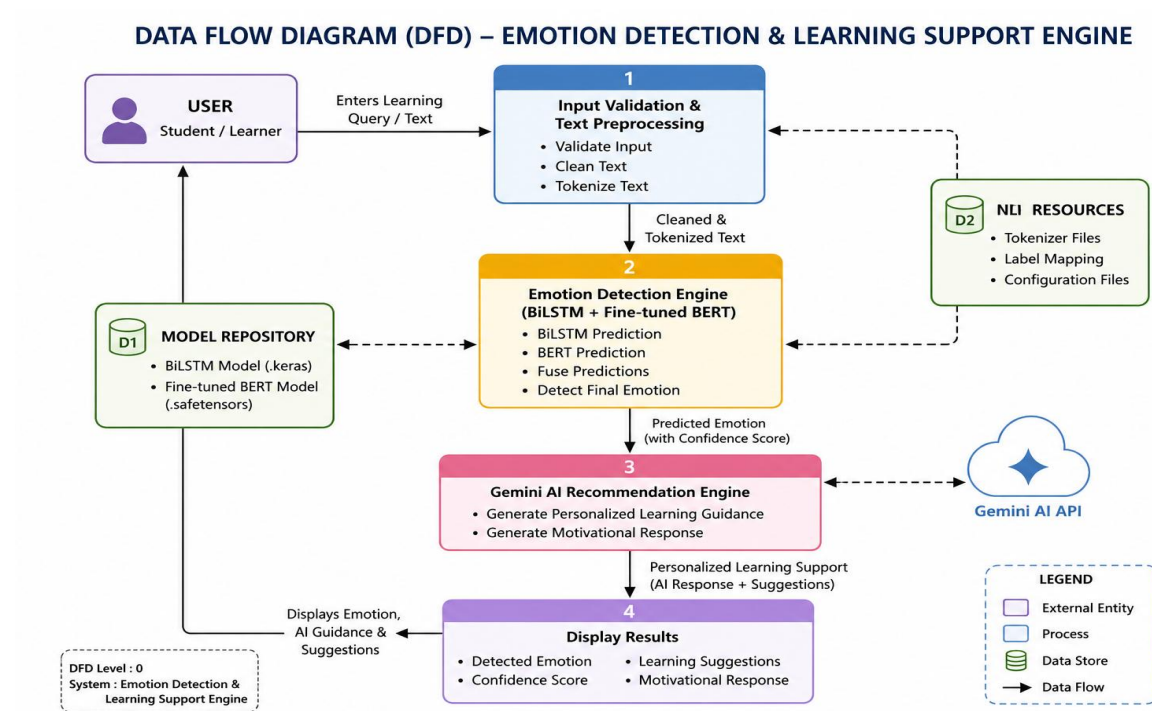
The Data Flow Diagram (DFD) illustrates how data flows through the **Emotion Detection & Learning Support Engine**. It shows the interaction between the user, the Streamlit web application, the preprocessing module, the emotion detection models (BiLSTM and BERT), the Gemini AI API, and the final personalized learning recommendations. The DFD represents how user inputs are processed to detect emotions and generate intelligent learning support.

### DFD Symbol Legend

| Symbol                         | Name            | Description                                                                                                                           |
|--------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Oval / Rounded shape           | External Entity | Represents an external user interacting with the system by providing learning-related text and receiving personalized recommendations |
| Rectangle with numbered header | Process         | Represents system processes such as text preprocessing, emotion detection, AI response generation, and recommendation generation.     |

|                                   |            |                                                                                                                              |
|-----------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------|
| Rectangle (solid fill, no number) | Data Store | Stores trained machine learning models, tokenizer files, configuration files, and prediction history used during processing. |
| Labeled arrow                     | Data Flow  | Represents the movement of data between users, system processes, machine learning models, APIs, and generated outputs.       |

## Data Flow Diagram:



## ER Diagram:

### ER DIAGRAM – EMOTION DETECTION & LEARNING SUPPORT ENGINE

